TEACHER PORTAL

School Management System

Technical Specification & Implementation Guide — v1.1 · March 2026
Confidential · Internal Documentation

Audience
Development Team · Product Team · School Administrators

Related Modules
Student Management · Academics · Attendance · Examination · E-Learning · Parent Module · Communication System · Timetable · Security · Staff Management

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1. MODULE OVERVIEW

The **Teacher Portal** serves as the operational workspace where teachers manage their daily academic responsibilities within the School Management System.

The portal is designed for **high-frequency usage**, meaning that performance, usability, and data accuracy are critical.

Primary objectives include:

- streamline classroom administration
- simplify attendance recording
- support assignment and grading workflows
- provide communication with parents and students
- enable resource sharing and lesson planning

The system must support **real-time data synchronization** with other modules to ensure consistency across the platform.

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2. CORE SUBMODULE STRUCTURE

Submodule	Description	Priority	Status
Teacher Dashboard	Daily overview and quick actions	Core	New
Class & Subject Management	Teacher class assignments	Core	New
Attendance Management	Record and manage attendance	Core	New
Assignment Management	Create and manage assignments	Core	New
Gradebook	Record student grades	Core	New
Examination Interface	Manage exam scores	Core	New
Learning Resources	Upload and share materials	Core	New
Communication	Messaging with parents and students	Core	New
Teacher Timetable	View teaching schedule	Core	New
Analytics & Reports	Class performance analysis	Enhanced	Optional
Lesson Planning	Organize teaching plans	Enhanced	Optional
Student Behavior	Record discipline or positive notes	Enhanced	Optional

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3. AUTHENTICATION & SECURITY

Teacher accounts are managed through the **Staff Management Module**.

Login Credentials

Field	Description
Username	Staff ID or Email
Password	System-generated temporary password

Example:

```
Username: teacher@school.edu
Password: ********
```

Authentication Flow

```
Login Page
    |
Teacher Authentication
    |
JWT Token Generated
    |
Redirect to Teacher Dashboard
```

Security Policies

Policy	Rule
Session timeout	30 minutes inactivity
Failed login limit	Lock after 5 attempts
Password reset	Email verification
Device logging	Record browser and IP

4. STAFF TENANT PASSWORD MANAGEMENT

Password management for staff accounts follows **administrator-controlled security policies**.

Account Creation

When a teacher account is created:

1. Administrator registers staff details.
2. System generates temporary password.
3. Credentials delivered via secure channel.

Example delivery:

```
Email
SMS
Secure onboarding slip
```

First Login Flow

```
Staff receives credentials
       |
Login with temporary password
       |
Forced password change
       |
Access granted to Teacher Portal
```

Password Reset

Administrators may initiate password resets.

Process:

```
Admin triggers reset
      |
System generates temporary password
      |
Staff receives reset notification
      |
Password must be changed at next login
```

Reset notifications may use:

- Twilio
- Africa's Talking

Password Security Rules

Requirement	Rule
Minimum length	8 characters
Uppercase	Required
Numbers	Required
Special characters	Required

Administrators cannot view staff passwords.

Only **reset capability** is permitted.

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5. DASHBOARD DESIGN

The **Teacher Dashboard** provides a centralized overview of daily teaching activities.

Dashboard Widgets

```
Teacher Dashboard
│
├ Today's Classes
├ Attendance Pending
├ Assignments To Grade
├ Recent Student Submissions
├ Parent Messages
└ School Announcements
```

Quick Actions Panel

Quick action buttons reduce navigation complexity.

```
Quick Actions
│
├ Take Attendance
├ Create Assignment
├ Enter Grades
├ Upload Resource
└ Send Announcement
```

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6. CLASS & SUBJECT MANAGEMENT

Teachers access only the classes and subjects assigned to them.

Class Structure

```
Subject
│
├ Class
│    ├ Student List
│    ├ Attendance
│    ├ Assignments
│    └ Grades
```

Example:

```
Mathematics
└ Grade 10A
```

Teachers may:

- view enrolled students
- review class performance
- access subject curriculum

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7. ATTENDANCE MANAGEMENT

Attendance recording must be optimized for **speed and simplicity**.

Attendance Interface

Example class attendance screen:

```
Grade 10A – Mathematics

✓ John Doe
✓ Mary Smith
✗ Ahmed Khan
✓ Jane Adams
```

Attendance Status Types

Status	Description
Present	Student attended
Absent	Student absent
Late	Student arrived late
Excused	Approved absence

Attendance Workflow

```
Teacher opens class
        |
System loads student list
        |
Teacher marks attendance
        |
Attendance saved automatically
```

System auto-records:

- date
- class
- subject
- teacher

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8. ASSIGNMENT MANAGEMENT

Teachers create and track assignments through the portal.

Assignment Features

Feature	Description
Assignment creation	Define homework or projects
File attachments	Upload supporting resources
Submission deadline	Define due date
Student submissions	Upload student work
Feedback	Teacher comments and grading

Assignment Workflow

```
Teacher creates assignment
        |
Students receive notification
        |
Students submit assignments
        |
Teacher reviews submissions
        |
Grades recorded
```

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9. GRADEBOOK SYSTEM

Teachers record grades for various assessments.

Grade Categories

Category	Example
Homework	Assignment marks
Quiz	Short tests
Exam	Midterm / Final
Participation	Class activity

Gradebook Example

```
Student        Assignment 1   Quiz 1   Midterm
-----------------------------------------------
John Doe             85%        90%      78%
Mary Smith           92%        88%      91%
```

The system calculates:

- subject averages
- class averages
- student ranking

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10. EXAMINATION INTERFACE

Teachers interact with the exam module to record results.

Exam Functions

Teachers can:

- view exam schedule
- enter exam scores
- review performance summaries

Example:

```
Exam: Mathematics Midterm

John Doe → 78
Mary Smith → 91
Ahmed Khan → 64
```

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11. LEARNING RESOURCE MANAGEMENT

Teachers upload materials for student access.

Supported Content Types

Type	Example
Documents	PDFs
Slides	PowerPoint
Videos	Recorded lessons
Links	External resources

Video content may be embedded from:

- YouTube

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12. COMMUNICATION SYSTEM

Teachers communicate with students and parents through the integrated communication module.

Capabilities include:

- responding to parent messages
- sending announcements
- scheduling meetings

Communication integrates with the **Parent–Teacher Communication System**.

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13. TEACHER TIMETABLE

Teachers access their teaching schedule through the portal.

Example Schedule

```
Monday

08:00 – Grade 10A Mathematics
09:00 – Grade 9B Mathematics
10:00 – Free Period
11:00 – Grade 11C Mathematics
```

Features include:

- daily timetable
- weekly view
- classroom locations

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14. ANALYTICS & REPORTING

Teachers can monitor class performance trends.

Analytics Metrics

Metric	Description
Class average	Overall performance
Attendance trends	Absence patterns
Top performers	High-performing students
Struggling students	Students needing intervention

Example:

```
Mathematics Class Average
78%
```

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15. SECURITY & PERMISSIONS

Access control ensures teachers interact only with authorized data.

Role Access

Role	Access Scope
Teacher	Assigned classes
Head Teacher	Department classes
Administrator	Full system access

Teachers cannot access:

- other teacher classes
- administrative financial data

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16. API ENDPOINT STRUCTURE

Teacher module APIs are organized under:

```
/api/teacher/
```

Core Endpoints

Method	Endpoint	Description
GET	/api/teacher/dashboard	Dashboard data
GET	/api/teacher/classes	Assigned classes
POST	/api/teacher/attendance	Submit attendance
POST	/api/teacher/assignment	Create assignment
POST	/api/teacher/grades	Record grades
GET	/api/teacher/timetable	Teacher schedule

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17. DATABASE STRUCTURE

Teacher Classes Table

Field	Type
id	UUID
teacher_id	FK
subject_id	FK
class_id	FK

Attendance Table

Field	Type
id	UUID
student_id	FK
class_id	FK
subject_id	FK
status	Enum
date	Date

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18. MODULE INTEGRATIONS

Module	Purpose
Student Management	Student records
Academics	Subject curriculum
Attendance	Attendance tracking
Examination	Exam management
Parent Module	Parent communication
Timetable	Teacher scheduling
Communication	Messaging system
Security	Access control

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19. IMPLEMENTATION WORKFLOW

Typical teacher workflow:

```
Teacher Login
      |
Open Dashboard
      |
View Today's Classes
      |
Take Attendance
      |
Teach Lesson
      |
Create Assignments
      |
Grade Submissions
      |
Respond to Parent Messages
```

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20. DESIGN PRINCIPLES

The Teacher Portal follows three core design principles:

```
Efficiency
Accuracy
Teaching Support
```

The portal must enable teachers to perform academic tasks quickly while minimizing administrative burden.